Statistical Analysis for a One Way Random Effects ANOVA

Personnel Officers Data Set

In Lecture Notes no 6 we have discussed the one way random effects ANOVA model and the data set for personnel officers. The SAS file and SAS output for this data set is posted on the Husky CT cite. In this data set, the experimenter has selected at random $a = 5$ personnel officers from large pool of personnel officers working for this company. $N = 20$ applicants of equal qualifications have been selected to this experiment and $n = 4$ applicants have been assigned at random to each personnel officer. This is a random effects model because the 5 personnel officers have been selected at random. The model for this experiment is:

$$Y_{ij} = \mu + \tau_i + \varepsilon_{ij}, \tag{1}$$

where Y_{ij} is the score assigned by the ith personnel officer to the jth applicant assigned to the ith personnel officer, τ_i is the random effect of the ith personnel officer on the assigned score, and ε_{ij} is the random error. In this model it is assumed that τ_i is a random variable with a normal distribution with mean = 0 and variance equal to σ_τ^2 and ε_{ij} is a random error with a normal distribution with mean = 0 and variance equal to σ_e^2. Based on that we get that the population mean of Y_{ij} is equal to μ. In this model we test the following null hypothesis:

$$H_0 : \sigma_\tau^2 = 0 \tag{2}$$

vs. the following alternative hypothesis:

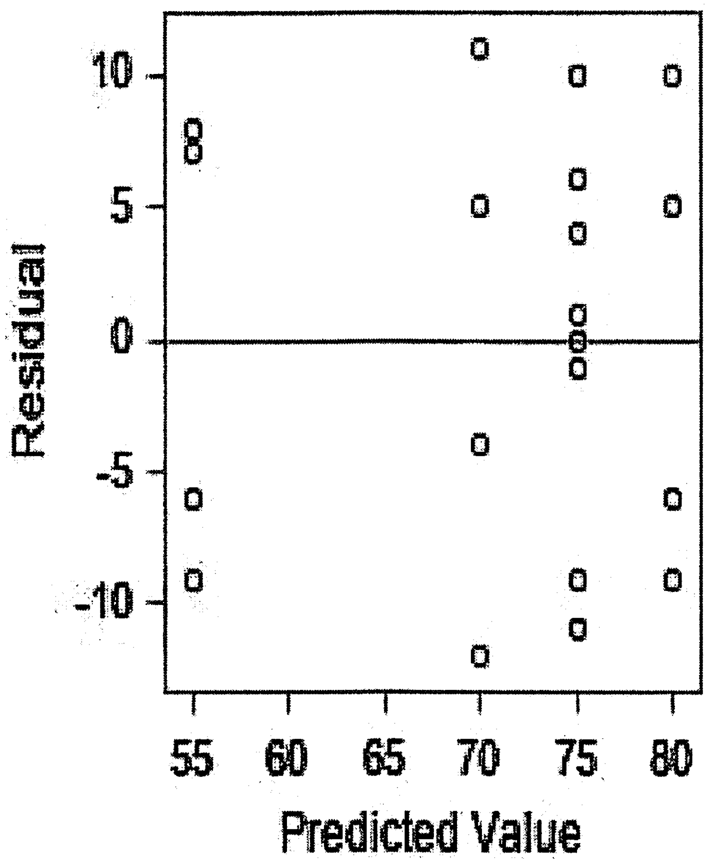
$$H_1 : \sigma_\tau^2 > 0.$$

From the SAS output we get the following ANOVA table that includes the F test and its p-value for testing the hypotheses listed above:

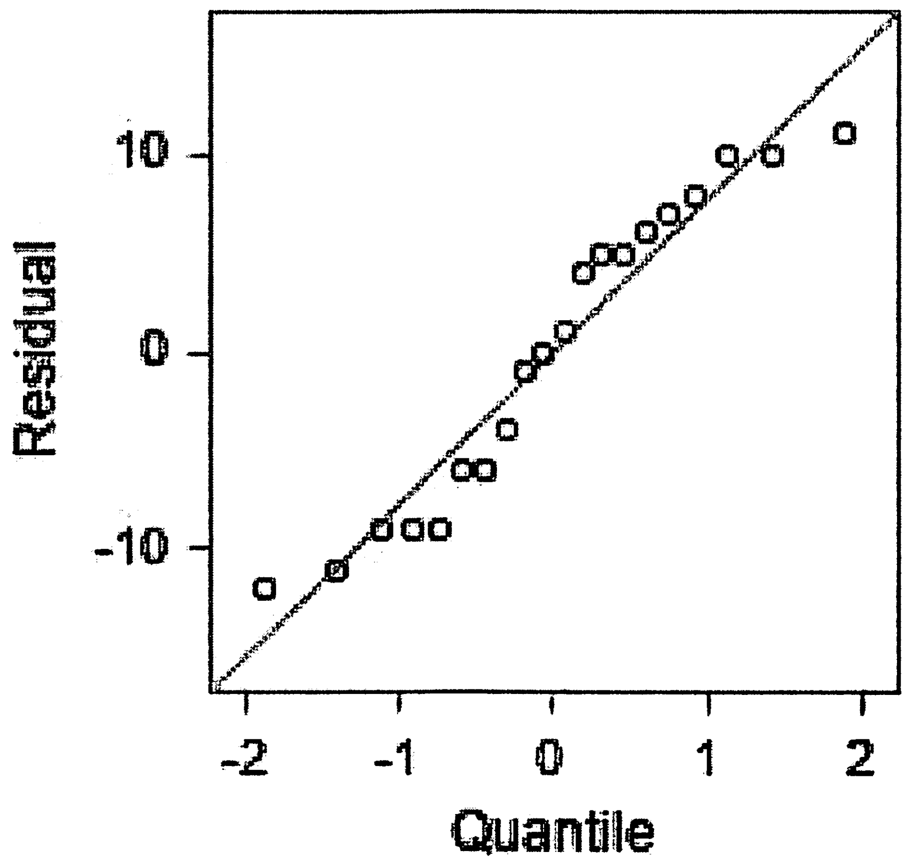
Dependent Variable: SCORE

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	4	1480.000000	370.000000	4.89	0.0100
Error	15	1134.000000	75.600000		
Corrected Total	19	2614.000000			

Based on this we can reject the null hypothesis at the .01 level. For this test and the conclusion we have made to be valid, one has to check the underlying assumption for our model in (1). For this we inspect the plots in the Fit Diagnostics panel that is on page 3 of the SAS output. First we examine the plot of the residuals vs. the predicted values:

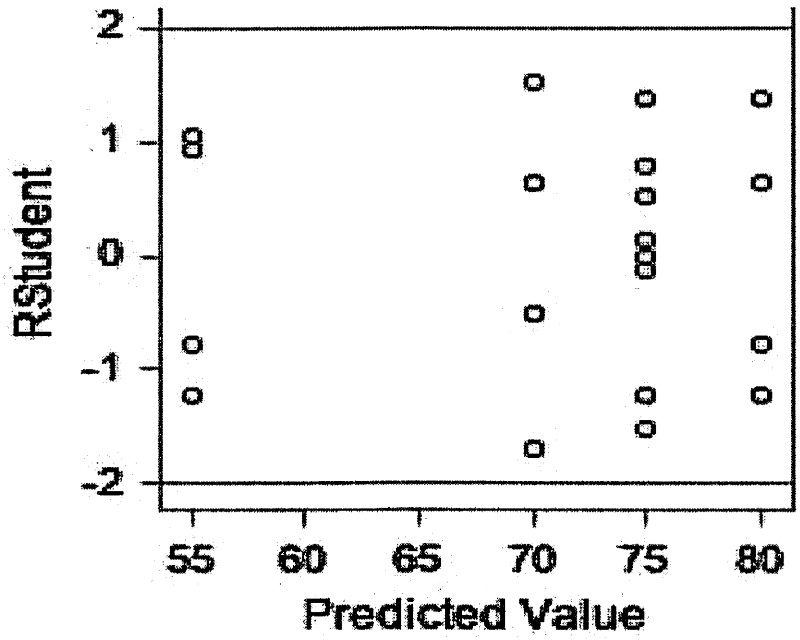


Note that the residuals are randomly scattered about the 0 line and there are no patterns. This validates the ANOVA model and the fact the variances of the random errors are constant. Next we examine the normal probability plot and the p-value of the W statistic.



Note that the normal probability looks pretty straight and the p-value of the W statistic, given in the Table below, is .1062 > .10. This supports out assumption that the random errors are normally distributed.

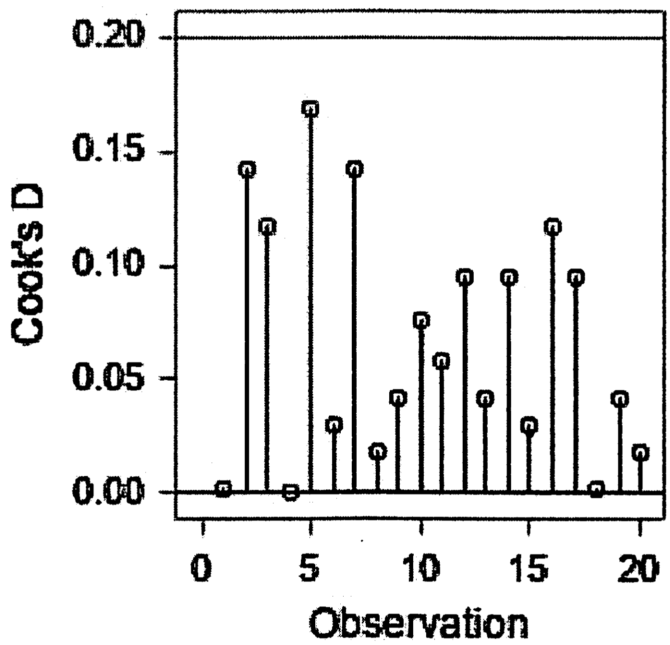
Tests for Normality				
Test	Statistic		p Value	
Shapiro-Wilk	W	0.921574	Pr < W	0.1062
Kolmogorov-Smirnov	D	0.147688	Pr > D	>0.1500
Cramer-von Mises	W-Sq	0.08301	Pr > W-Sq	0.1848
Anderson-Darling	A-Sq	0.539478	Pr > A-Sq	0.1488



Next we examine two plots: the RStudent values vs. Predicted values, for potential outliers, and the Cook's D values plot for potential influential observations.

From the plot above, since all RStudent values are less than 2 in absolute values (it would have sufficient they were less than 4 in absolute value) we can conclude that this data set has no outliers.

From the plot below, we see that all Cook's D value are less than 1, in fact they are less than .20, which implies that there are no influential observations in this data set.


0.20
0.15
0.10
0.05
0.00
Cook's D
0
5
10
15
20
Observation

Based on the information above we can conclude that all the underlying assumption for our model are valid. Since, the null hypothesis in (2) has been rejected, we have to estimate the variance components in our model. From the ANOVA table given above we get that:

$$\widehat{\sigma_e^2} = MSE = 75.6.$$

To estimate the variance component due to treatment,.i.e. due to personnel officers:

$$\widehat{\sigma_\tau^2} = \frac{MS_{Treatment} - MSE}{n} = \frac{370 - 75.6}{4} = 73.6.$$

Note that in the SAS output, for a one way ANOVA

$$MS_{Treatment} = MS_{Model}.$$

Therefore, the total variability in the data is:

$$\widehat{\sigma_{Total}^2} = \widehat{\sigma_\tau^2} + \widehat{\sigma_e^2} = 73.6 + 75.6 = 149.2.$$

Based on that, the proportion of variability due to the personnel officers is:

$$\rho = \frac{73.6}{149.2} = .4933.$$

This means that about 49% of the variability in the data is due to personnel officers. As we have conclude in class, the personnel officers in this company will have to be retrained, so that the variability is reduced.